MIL-T-62130B(AT)
27 January 1976
SUPERSEDING
MIL-T-62130A(AT)
1 March 1971

MILITARY SPECIFICATION

TRUCK, TANK: FUEL SERVICING 2500 GALLON, 4X4, M559

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a high mobility (GOER type), 2500-gallon tank capacity bulk fuel hauler, diesel engine driven, two or four-wheel optional drive military truck. This vehicle shall operate cross-country, swim inland waterways, and be air transportable.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bid or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

VV-F-800	-	Fuel Oil, Diesel.

Military

MIL-H-370	-	Hose and Hose Assemblies, Rubber, Liquid, Fuel, Non-Collapsible.
MIL-P-514	-	Plates, Identification, Instruction and Marking, Blank.
MIL-G-3056	-	Gasoline, Automotive, Combat.
MIL-R-3065	-	Rubber, Fabricated Products.
MIL-A-8421	-	Air Transportability Requirements, General Specification for.
MIL-F-8901	-	Filter-Separators, Aviation and Motor Fuel, Ground and Shipboard Use, General Requirements and Test Procedures for.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

MIL-P-17802 - Padlocks, and Padlock Sets, Pin Tumbler Mechanism, Key Operated.
MIL-A-46153 - Antifreeze, Ethylene Glycol, Inhibited, Heavy-Duty, Single Package.
MIL-V-62038 - Vehicles, Wheeled: Preparation for Shipment and Storage of.
MIL-T-62129 - Tire, Pneumatic: For Truck, Logistical GOER Type, (Tubeless).
MIL-E-62133 - Engine, Diesel: 6 Cylinder, In-Line, 213 H.P. Turbo-Supercharged.
MIL-T-62134 - Transmission, Power Shift, Commercial Modified.

STANDARDS

Federal

FED. STD. NO. 595 - Colors.

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection By Attributes.
MIL-STD-130 - Identification Marking of U. S. Military Property.
MIL-STD-193 - Painting Procedures, Tactical Vehicles, (Tracked and Wheeled).
MIL-STD-210 - Climatic Extremes for Military Equipment.
MIL-STD-417 - Rubber Compositions, Vulcanized General Purpose, Solid (Symbols and Tests).
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-642 - Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-882 - System Safety Program for Systems and Associated Subsystems and Equipment: Requirements for.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474 - Noise Limits for Army Materiel
MS-51326-1 - Heater, Vehicular Compartment: Liquid Coolant, 30,000 BTU/HR.
MS-53063 - Indicator, Air Cleaner, Intake, Restriction Mechanical Type.

DRAWINGS

11614319 - Truck, Tank: Fuel Servicing, 2,500 Gal. 4x4 M559.
11614953 - Mounting Kit, Machine Gun, Cal. 50.
11614988 - Arctic Kit.
11618590 - Meter Kit Installation.
11624590 - Frame and Canopy Kit Installation.
11624639 - Filter Kit.
11651365 - Light Kit, Rotating Warning.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:23Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction vehicle. When specified (see 6.2), a complete preproduction vehicle(s) to be furnished un der this specification shall be produced prior to the manufacture or fabrication of vehicles in quantity. The vehicle(s) when complete shall be submitted to the Government for examination and tests to determine conformance to the requirements of this specification. Vehicle(s) submitted by the contractor shall be fully representative of vehicles proposed to be supplied by the vehicle manufacturer from production facilities and tooling under the supply contract.

3.2 Initial production vehicle(s). The vehicle(s) selected shall be fully representative of vehicles proposed to be furnished under the contract and shall be submitted to the Government for inspection to determine conformance to the requirements of this specification.

3.2.1 Reliability (specified). During the first 18,000 miles of operation, the vehicle shall have a Mean Miles Between Mission Failure (MMBMF) of 1,000 miles. A mission failure is defined as a malfunction which:

(a) Will immobilize the vehicle, produce a critical or catastrophic safety hazard as defined by MIL-STD-882.
(b) Cannot be corrected within one hour (including diagnostic time) by the crew using On Equipment Material (OEM).

3.2.1.1 Vehicle mission. Reliability operation shall be conducted in the ratio of mileage and requirements of the mission statement (see 6.5).

3.2.2 Durability (minimum). During the first 18,000 miles of vehicle operation, the engine and transmission shall have a combined probability of 65% of not requiring replacement.

3.2.3 Maintainability. When maintenance procedures specified in the technical manuals are followed and the vehicle is operated as defined in the mission statement, the demonstrated maintainability shall be as follows:

(a) Mean Operator Preventive Maintenance Time shall not exceed 1.50 manhours per mission. This time shall not be included in organizational preventive maintenance time.
(b) Maximum Operator Corrective Maintenance Time shall not exceed 1.0 manhours per mission without being classified as a mission failure.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

(c) The ratio of total corrective and organizational preventive maintenance manhours to operating hours shall not exceed .36.
(d) The ratio of total organizational preventive maintenance manhours to total operating hours shall not exceed .11.
(e) The ratio of total corrective maintenance manhours to operating hours shall not exceed .25.
(f) Mean manhours to perform a corrective maintenance action (MTTR) shall not exceed 5.00.
(g) The Mean Time Between Corrective Maintenance Actions (MTBF) shall not be less than 20.0 operating hours.
(h) Inherent Availability (Ai) shall be not less than .89.

$$Ai = \frac{MTBF}{MTBF + MTTR}$$

NOTE: In the above formula, MTBF is expressed in engine hours and MTTR is expressed in clock hours. Where specific hours are not available, appropriate conversion factors (i. e., to change from man hours of maintenance to clock hours of down time) should be used.

3.3 Materials.

3.3.1 Qualified products list (QPL). The contractor shall be responsible for using all assemblies and parts necessary in vehicle construction that are listed as qualified products on the QPL and those approved for inclusion on the QPL. Assemblies and parts determined as QPL items appear on the Engineering List of Specifications cited in 3.4. The contractors manufacturing inspection records shall specifically list all QPL items by part or drawing number(s), name of supplier(s), and number and date of the QPL. When assemblies and parts are approved as qualified products but not as yet listed on the QPL, the contractor shall list the products by number and date of the approval document with name of supplier(s).

3.3.2 Ozone resistant products. Rubber products procured under requirements of MIL-R-3065 are to be ozone resistant to the climatic environment specified herein and by the requirements of MIL-STD-417. Rubber components controlled by detail specifications shall be ozone resistant only when required by the specification referenced on the drawing.

3.4 Contruction. Vehicles, components, subassemblies, and assemblies shall be fabricated and assembled into a complete vehicle in accordance with 11614319 and associated lists. All parts, subassemblies, and assemblies shall be identified in accordance with MIL-STD-130 when specified on drawings.

3.4.1 Seals.

3.4.1.1 Bearing and lubricant seals. With the vehicle operating under all types of cross-country terrain and during fordings or swimming, the seals

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

shall restrict the entrance of all foreign matter into the bearings, which are exposed to contamination during these operations. Seals shall prevent water contamination of lubricants, and water contamination shall not exceed 2 percent by volume. All bearing seals shall prevent the leakage of lubricants from the bearings.

3.4.2 Special kits. When specified (see 6.2 and 6.4), special kits shall be furnished and shall meet the requirements specified herein.

3.4.2.1 Canopy and frame kit. The canopy and frame kit shall conform to 11624590.

3.4.2.2 Arctic kit. The arctic kit conforming to 11614988 shall provide a hard top cab for the tractor, heat for the cab, and assure engine starting and operation in ambient temperatures from minus 25°F (4 to 6 feet above ground) to minus 65°F (measured at ground level).

3.4.2.3 Meter kit fuel dispensing. The meter kit, when installed, shall meter at a rate of not less than 50 gallons per minute (gpm) on each of the 1 1/2 inch hose reel, 100 gpm on the 2 inch hose reel and 300 gpm on the 3 inch outlet.

3.4.2.4 Infra-red filter kit. The infra-red filter kit conforming to 11624639 shall be installed.

3.4.2.5 Gun-ring mounting kit. The gun-ring mounting kit conforming to 11614953 when installed shall permit 360 degrees rotation without interference.

3.4.2.6 Warning light kit. The warning light kit shall conform to 11651365.

3.4.3 Tire chains. The tire chains, when installed, shall retain maximum tightness at tire manufacturer's rated pressure to prevent damage to wheel walls and vehicle underbody. Scuffing of body metal shall be considered as damage.

3.4.4 Power steering unit. The power steering unit shall operate without malfunction or leakage of lubricants under all vehicle operating conditions.

3.4.5 Engine. The vehicle performance requirements shall be met when engine conforming to MIL-E-62133 is properly installed in vehicle.

3.4.5.1 Engine air induction system. The engine air induction system, as fabricated and properly installed from air cleaner to engine turbo charger intake, shall be free from all weldslag and foreign material. The system shall prevent entrance of all foreign material and withstand pressure up to 5 psi. Air cleaner restriction indicator conforming to MS-53063-5 shall be installed.

3.4.6 Transmission. The vehicle performance requirements shall be met when transmission conforming to MIL-T-62134 is properly installed in vehicle.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

3.4.7 Exhaust system. The exhaust system as installed shall be gastight and leakproof such that during normal vehicle operation the concentration of NO_x in the crew compartment shall not exceed 5ppm and the concentration of carbon monoxide shall not exceed 50 ppm.

3.4.8 Fuel tanks and hydraulic reservoir. Provisions shall be made to assure the internal cleanliness of the fuel tanks and hydraulic reservoir prior to initial filling.

3.4.9 Engine stop switch. When the engine stop switch is activated, the engine shall stop within 2 minutes.

3.4.10 Fuel, air, and hydraulic lines, and fittings. All fuel air, hydraulic lines, and fittings shall be internally clean prior to and after making connections and shall be free from leaks. All lines and fittings shall be secured in such a manner to prevent rubbing on adjacent lines or vehicle appendages.

3.4.11 Governor control linkage. Governor control linkage (throttle) shall be adjusted to permit the full range of engine-governed RPM capability. The accelerator pedal travel, when fully depressed, shall be restricted by adjustment to preclude damage to governor and governor control linkage.

3.4.12 Personnel heater (hot water type). The personnel heater MS51326-1 as installed shall maintain temperatures of not less than 40°F within the vehicle cab in ambient air temperatures down to minus 25°F and provide defrosting of the windshield.

3.4.13 Front winch. When specified (see 6.2), the front mounted winch as installed shall have a single line capability of not less than 10,000 pounds on the bare drum. The wire rope for the winch shall be not less than 150 feet in continuous length, without splice or joint and shall be provided with a chain and hook.

3.4.14 Tank shell. The tank shell after all welded components are affixed, manhole and filler covers are closed and locked, shall not leak or permanently deform when subjected to air or hydrostatic pressure of not less than 3 psig.

3.4.14.1 Emergency valve. The emergency valve assembly shall provide the means whereby fuel is permitted to flow into the equipment compartment plumbing. The valve is to be opened by hydraulic pressure (150 psi maximum) hand pumped. The valve shall close or shut off in three (3) different places as follows:

(a) Lower the lever on the hand pump and press in the knob to lock in release position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

(b) Depress release lever by closing hatch door thereby permitting the 30 psi pressure to drop and close emergency valve.
(c) Pull the remote control lever down, which is located on the left front of tanker body releasing the hydraulic pressure and close valve. The hand operated valve control shall contain a fusible section causing the valve to close in case of fire.

3.4.14.2 Piping and accessories. All tank piping and fuel dispensing pump and couplings, when installed, shall not leak statically or under pressure.

3.4.14.3 Fuel dispensing pump and filter separator. The fuel dispensing pump shall have a capacity of not less than 300 gpm at hose discharge end when using 3 inch hose with applicable coupling at a main engine speed of 1700 rpm.

3.4.14.3.1 Valving (fuel dispensing pump). The valving for the fuel dispensing pump shall permit bottom filling and bottom discharge of fuel at the rate of not less than 300 gpm and shall also permit gravity discharge of fuel.

3.4.14.4 Reels and hoses.

3.4.14.4.1 Reels and discharge hoses. The 25 foot sections of discharge hose furnished with the reels shall conform to type I of MIL-H-370. With applicable quick disconnect couplings nozzle and adapters, the 1 1/2 inch hose furnished on 2 reels and 2 inch hose furnished on 1 reel shall permit discharging of fuel at the rate of 50 gpm for the 1 1/2 hose and 100 gpm for the 2 inch hose.

3.4.14.4.2 Suction hose. The two 3 inch suction hose assemblies 15 feet in length with applicable couplings shall permit a delivery capacity of 300 gpm minimum. The suction hose shall conform to type II of MIL-H-370.

3.4.14.4.3 Grounding wire. The grounding wire shall be provided and be easily accessible in the general area of the hose reel opening. The wire shall be mounted on a spring retractable reel.

3.4.14.5 Equipment operator's cabinet. The hose and reel doors of the operator's cabinet shall have a locking mechanism and shall open and close with ease and without interference. The doors when closed and locked shall seal out mud, snow, rain and shall restrict water from entering the cabinet during swimming operations.

3.4.15 Controls. Electrical, mechanical, pneumatic and hydraulic controls shall operate without malfunction throughout all ranges of operation under all vehicle operating conditions.

3.4.16 Adjustment mechanisms. Adjustment mechanisms shall function properly and maintain adjusted settings during all conditions of vehicle operations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

3.4.17 Stowage. Stowage provision made shall have the capability to contain the manufacturer's, depot and troop installed Equipment List (EL). Padlocks as required shall conform to class II of MIL-P-17802.

3.4.18 Welding repairs and burning operations. Welding repairs and burning operations of any type or class shall be made only when, and to the extent specifically authorized by procuring activity.

3.4.19 Transportability. The vehicle shall be transportable by ship and landing vessels and transportable by railways without major disassembly. Air transportability shall be for type C141 and C5A aircrafts and shall meet requirements of MIL-A-8421.

3.4.20 Tires. Vehicle equipped with tires conforming to MIL-T-62129 shall meet all requirements specified herein.

3.4.21 Soft top. Unless otherwise specified (see 6.2), the contractor shall furnish the soft top enclosure for the crew compartment.

3.4.22 Safety. The vehicle shall be evaluated in accordance with MIL-STD-882 to assure all hazards have been eliminated or reduced to an acceptable level through the utilization of appropriate safety devices.

3.5 Break-in run. Prior to break-in run, each vehicle produced shall be completely assembled and all adjustments completed (except for equipment, component installation and servicing performed by the contractor as a final processing of vehicles for acceptance). Vehicle, including all components requiring lubrication, shall be lubricated. Vehicle shall be serviced with fuel conforming to VV-F-800. Lubricants and fluids shall be as specified in lubrication chart. Specific applications by grades or types for various expected seasonal temperature ranges shall be as specified therein.

3.5.1 Oil pressure. Prior to start of the break-in run, the engine shall be operated at 700 to 800 RPM until lubricating oil is at operating pressure in accordance with MIL-E-62133. If at any time during the break-in run, the engine has been stopped for at least 15 minutes, the engine shall again be operated as above before continuing the break-in run. Proper oil pressure shall be maintained during operation on level ground, 60% grades and 30% side slopes.

3.5.2 Distance. Each vehicle shall be operated by the contractor for a distance of 15 miles consisting of the following types of operation:

(a) 1st gear, 1/2 mile
(b) 2nd gear, 1 mile
(c) 3rd gear, 1 1/2 miles
(d) 4th gear, 1 1/2 miles
(e) 5th gear, 4 1/2 miles
(f) 6th gear, 6 miles

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

NOTE: During the above runs, the vehicle shall be stopped and then operated in reverse gear 3 times, 100 feet each time.

3.5.2.1 Condition after run. After completion of the break-in run, the engine shall run smoothly. Prior to vehicle being submitted to the Government for acceptance, the vehicle shall be checked to determine if adjustments are correct. There shall be no defects indicative of damaged components that may cause faulty vehicle operation.

3.6 Performance. The complete vehicle, serviced and equipped for existing environmental and climatic condition and carrying full payload of 16,000 pounds shall meet performance requirements specified herein.

3.6.1 Environmental. The vehicle shall operate in ambient air temperature of plus 125°F (ground surface temperature) to minus 25°F (4 to 6 feet above ground) to minus 65°F (ground surface temperature). The complete vehicle, when in storage shall withstand climatic extremes as specified in MIL-STD-210, without deterioration that may cause failure of any component part of vehicle.

3.6.2 Grade and slope operation. The vehicle with full payload (without towed load) shall operate in forward and reverse gears on longitudinal grades up to 60% and side slopes up to 20% (30% only with extreme caution) with each side of the vehicle up slope. As a result of the operation, no evidence of faulty lubrication, cooling, fuel supply, leakage or other malfunction shall be found.

3.6.3 Engine starting on grades and side slopes. With the vehicle standing on a 60% grade, for not less than 2 minutes with engine operating under no load between 800 to 1,000 rpm, the engine shall be stopped for no less than 2 minutes. The engine shall restart in not more than 2 minutes when vehicle is headed up and headed down grade respectively. A similiar engine starting test shall be conducted on 20% side slopes with right side and then left side of vehicle up slope.

3.6.4 Cooling system. The engine coolant system and engine and transmission coolers as installed shall control the coolant and oil temperatures, to prevent damage to the engine and transmissions, under all conditions of vehicle operation. Temperatures shall not exceed those shown in Table I when vehicle is operated in ambient temperatures of / 125°F.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:23Z
Check the source to verify that this is the current version before use.

Table I. Operating temperatures.

Maximum engine coolant temperature at engine outlet.	Maximum engine oil temperature measured at oil sump drain plug.	Maximum transmission oil temperature measured at converter outlet.
230°F.	250°F.	275°F.

3.6.5 Cruising range. Vehicle with payload shall operate for not less than 300 miles at average speed of 20 miles per hour on hard-surfaced roads over average rolling terrain without refueling.

3.6.6 Speeds. With rated payload, but less towed load, the vehicle shall maintain a speed of not less than 30 mph on hard-surfaced roads with a grade not greater than ± 1/2 percent, and in low gear maintain a minimum speed of 2 1/2 miles per hour.

3.6.6.1 Towed loads. The vehicle loaded with its rated payload shall be capable of towing loads up to 20,000 pounds GVW. During towed load operations, the vehicle, with payload, shall be operated at not less than 20 mph on hard-surface road with a grade not greater than ± 1/2 percent.

3.6.7 Braking ability.

3.6.7.1 Service brakes. When loaded with rated payload, the service brakes shall stop vehicle from a speed of 20 mph in not more than 30 feet from point of brake application. Deceleration and stopping distance shall be demonstrated on dry, brushed, concrete level road surface, free from loose material with a grade not greater than ± 1/2 percent. The service brake shall hold vehicle on a 60 percent grade when ascending and descending.

3.6.7.2 Parking brake. The parking brake shall hold vehicle loaded with rated payload headed up and headed down on grades of not less than 40 percent.

3.6.8 Steering angle. The power steering mechanism shall be adjusted to permit a turning angle of 60 degrees ± 2 degrees for both left and right turns. These turns shall be made with maximum articulation (roll) between the front and rear units, to either side without interference.

3.6.9 Vehicle articulation. Lateral articulation between the front and rear units shall be 20 degrees ± 2 degrees on each side of the longitudinal centerline. Stops, when used, shall limit the articulation of 5 degrees ± 3 degrees between the power unit and trailing unit to improve stability during lift and tow operations and during loading for shipment of vehicle by sea, land, and air.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:23Z
Check the source to verify that this is the current version before use.

3.6.10 Leakage during swimming. The vehicle with and without rated payload shall swim for a period of not less than 20 minutes. Leakage of water shall not exceed 1 quart per minute, total in tractor and carrier combined.

3.6.10.1 Bilge pumps. The power unit and trailing unit bilge pumps shall remove water from their specific areas at a minimum pumping rate of 46 gpm as installed.

3.6.11 Electrical system.

3.6.11.1 Voltage alternator-generator. The alternator voltage shall be nominally set at 28 volts at 2,000 rpm with 25 ampere load. The voltage spread shall not exceed ± .5 volt under all vehicle operating conditions, providing the electrical load does not exceed alternator capacity of 60 amps. The voltage shall be measured at the point of battery connection to the vehicle electrical system.

3.6.11.2 Lights and accessories. The internal and external lights, including cables and receptacles, shall function throughout all vehicle operating conditions. All electrical contacts and connections shall maintain positive contact under all vehicle operating conditions.

3.6.12 Fuel delivery system cleanliness. The fuel delivery system from the filter/separator outlet to either nozzle and fueling stub shall be cleaned to assure delivery of fuel at the nozzle or stub of the same quality as that at the filter/separator outlet in conformance to MIL-F-8901.

3.6.12.1 Cleaning. Prior to filter/separator installation, the tank and the complete fuel delivery system, including fuel dispensing hoses, shall be cleaned and flushed of all contaminates which may have accumulated during fabrication. After installation of filter/separator, extra care shall be taken to assure no water or other cleaning and flushing agent shall be introduced into the new filter/separator complete with unused elements properly installed.

3.6.12.2 Filter separator flow. The unit as installed shall not inhibit a rated flow of 300 gpm for type of fuel specified in VV-F-800, and MIL-G-3056. Filter performance shall be in accordance with MIL-F-8901.

3.7 Radio interference suppression. The complete vehicle shall be radio interference suppressed in accordance with tactical vehicle requirements class IIIA of MIL-STD-461.

3.8 Painting, marking and data plates.

3.8.1 Painting. The exterior and interior of the vehicle, components, assemblies, subassemblies, and parts that require painting shall be prepared and painted in accordance with MIL-STD-193. Vehicle exterior paint and exterior non-skid paint shall match color chip X24087 of FED. STD. No. 595.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:23Z
Check the source to verify that this is the current version before use.

3.8.2 Marking. Marking of vehicle shall be in accordance with drawings and shall conform to MIL-STD-642.

3.8.3 Name, shipping, and identification data plates. All plates shall be in accordance with drawings and shall conform to MIL-P-514.

3.9 Servicing and adjustment. Prior to delivery contractor shall service and adjust each vehicle and its mounted material for operational use including the following: the focusing of lights, the proper adjustment of the engine, running gear, adjustment of brake system, inflation of all tires, complete check of electrical, air and brake systems for proper operation, complete lubrication of chassis, engine, transmission, transfer case, steering gear, winches, axles and all running gear with proper grade of lubricant (per approved lubrication chart) for the climatic condition at the delivery point. Engine cooling system shall be serviced with a solution of ethylene glycol conforming to type I of MIL-A-46153 and water in equal parts by volume.

3.10 Noise and vibration levels. The noise and vibration levels associated with the vehicle and its elements in required combinationa, shall be controlled under all conditions and usage, to levels of tolerance of personnel in accordance with MIL-STD-1472 and MIL-STD-1474.

3.11 Workmanship. The vehicle shall be constructed and finished in a thorough workmanlike manner. Particular attention shall be paid to neatness and thoroughness of marking the parts, plating, painting, machine-screw assemblage, welding and brazing, and freedom of parts from burrs and sharp edges. Electric, hydraulic and lube lines shall be located in a manner so as to prevent possible damage by rubbing on adjacent lines or appendages. No condition shall exist which may create a safety hazard to operating or maintenance personnel.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's inspection or quality control system. The contractor shall provide and maintain an effective inspection or quality assurance system acceptable to the Government covering the supplies under the contract. A current, written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor shall notify the Government of any subsequent change to the submitted procedures prior to implementing the change into the system.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:23Z
Check the source to verify that this is the current version before use.

4.1.2 Government verification. All quality assurance operations performedy the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that practices, methods and procedures of the written inspection plan are being properly applied, and (b) Government inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed upon procedures or instances of poor practices which might have an effect upon the quality of the product will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct defects discovered shall be cause for suspension of acceptance until corrective action has been taken, or until conformance of product to specified criteria has been demonstrated.

4.1.3 Materials and qualified products. The contractor shall provide the Government with evidence that materials (see 3.3, 3.3.1, and 3.3.2) used are as specified in the applicable specifications and drawings.

4.1.4 Parts and components. Parts, components and assemblies shall be inspected in accordance with TDPL-11614319.

4.2 Preproduction and initial production inspection.

4.2.1 Preproduction inspection. A preproduction vehicle (see 3.1), when furnished, shall be inspected to ascertain vehicle conformance with the requirements of this specification. Inspection shall be performed at a location approved by the Government. Failure of the preproduction model(s) to comply with any of the requirements specified shall render it unacceptable to the Government until corrective measures satisfactory to the Government have been taken, and a successful retest has been conducted.

4.2.2 Initial production vehicle inspection.

4.2.2.1 Reliability test. Reliability requirement of 3.2.1 shall be verified while the vehicle is subjected to production (comparison) tests. The Minimum Acceptable Value (MAV) shall be not less than 725 Mean Miles Between Mission Failures (MMBMF). This value will be demonstrated at a level of confidence appropriate to the duration of the test.

4.2.2.2 Durability test. Durability requirement of 3.2.2 will be verified while vehicle is subjected to initial production tests.

4.2.2.3 Maintainability. The maintainability of the vehicle as specified in 3.2.3 will be verified during initial production testing.

4.2.2.4 In process inspection. During fabrication of the initial production vehicle (see 3.2), an in-process inspection will be conducted by a Government representative to evaluate conformance of materials and workmanship to requirements of specified documents. These inspections shall be made at the

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:23Z
Check the source to verify that this is the current version before use.

contractor's facility prior to the application of primer and paint. Processing, and welding procedures, quality system and inspection records will be reviewed and evaluated during this inspection.

4.2.2.5 Completed first vehicle inspection. The first production vehicle on each contract shall be road tested and completely inspected by the contractor for conformance to the requirements of contract and specifications. The applicable road test shall be in accordance with Paragraph 4.4.2.1. Upon completion, the contractor shall submit the vehicle and make available all inspection records and certifications to the responsible government inspection element at contractor's plant for preliminary inspection.

4.2.2.5.1 Preliminary inspection. The vehicle shall receive a preliminary inspection by the responsible government inspection element. The preliminary inspection shall be completed except for road test requirements of paragraph 4.4.2.1 which will be conducted jointly during provisional inspection to be accomplished by representatives of USATACOM Quality Assurance Directorate as specified in Paragraph 4.2.2.5.2.

4.2.2.5.2 Provisional inspection. The first completed production vehicle shall be subject to provisional inspection at the contractor's plant by representatives of USATACOM Quality Assurance Directorate. The contractor shall further provide any other inspection assistance as may be required. At the time of this inspection, the contractor shall make available his inspection plan, inspection records, and certifications pertinent to the vehicle and components.

4.2.2.6 Repairs of defects. Defects found as a result of above inspections shall be corrected by the contractor at no cost to the Government. Failure of the contractor to promptly correct defects shall be cause for suspension of acceptance of vehicles until corrective action has been accomplished.

4.2.2.7 Vehicle disposition. Upon completion of inspection and acceptance of the preproduction or first production vehicle, the vehicle shall remain at the manufacturing facility as a production sample. The contractor shall service and maintain the vehicle during this period in accordance with the applicable documents for care and preservation while in storage. At the discretion of the Purchasing Contracting Officer, the vehicle may be released for shipment.

4.2.2.8 Initial production test. Unless otherwise specified (see 6.2), to determine conformance to 3.2 through 3.2.3, the quality of vehicles specified in contract will be selected by the Government and shall be subjected to examinations specified in Table III, in addition to road tests specified in Table II and all tests specified in Table IV. The contractor shall support the test by providing any vehicle spare parts necessary for completion of such a test. Delay in test completion caused by vehicle breakdown or failure of the contractor to adequately support the vehicle with parts during the tests shall not be the basis for adjustment of the contract delivery schedule or contract price.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

Table II. 18,000 Mile Test

Course	Mileage & speeds	Vehicle payload (see 3.6)
Hard surface	3,600 miles paved at varying speeds up to maximum.	payload
Secondary roads	7,200 miles at speeds applicable to conditions of terrain.	payload
Hilly cross-country	3,600 miles at speeds applicable to conditions of terrain.	payload
Level cross-country	3,600 miles at speeds applicable to conditions of terrain.	payload
Swimming operation	8 hours of swimming in calm inland waters.	payload

NOTE: Ten percent of the mileage specified for each course shall be performed with towed load (see 3.6.6) swimming excepted.

4.2.2.8.1 Test failure. Failure of the test vehicle to meet minimum performance requirements, failure to comply with the requirements of a workmanship or materials nature and failure to manufacture the vehicle in accordance with the technical data revealed during or as a result of this test, shall be cause for rejection of the vehicle. Further, the Government may refuse to continue acceptance of production vehicles until evidence has been provided by the contractor that corrective action has been taken to eliminate the defects. Any defects found during or as a result of this test shall be prima facie evidence that all vehicles already accepted prior to completion of test, are similarly defective unless evidence satisfactory to the Government is furnished by the contractor that they are not similarly deficient. Such defects on all vehicles shall be corrected by the contractor at no cost to the Government.

4.3 Inspection provisions.

4.3.1 Lot size. An inspection lot shall consist of all vehicles of one type from one day's production, submitted at one time for acceptance inspection.

4.3.1.1 Sampling for inspection. For the purpose of visual, dimensional and primary functional inspection, a representative sample shall be selected from each inspection lot in accordance with inspection level II of MIL-STD-105. Before sampling may be accomplished by the contractor, the contractor shall 100% inspect the first 20 vehicles to establish a process average (see 6.7) to allow normal sampling in accordance with MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

4.3.1.2 Examinations. Visual, dimensional, and primary functional examination shall consist of examination of the vehicle for conformance to the applicable drawings and this specification. Examination shall be performed against the classification of defects and with the acceptable quality levels shown in Table III. The following constitutes a part of the classification of defects (see 6.3 and 6.8).

(a) Any drip constitutes a major defect when the vehicle has been standing idle and the components are at ambient temperature.
(b) Any droplet that occurs at a static fit metal to metal or gasket combination shall constitute a major defect.

Table III. Classification of Quality Defects

Defect No.	Defects	Method of Inspection
Major	A.Q.L. = 25 Defects/100 Units	
101	Steering mechanism: malfunction; unusual noise leaks (see 6.3)	Visual and functional
102	Engine: malfunction; unusual noise; leaks mounting improper	Visual and functional
103	Clutch: malfunction; unusual noise	Visual and functional
104	Transmission: malfunction; unusual noise; leaks (see 6.3) mounting improper	Visual and functional
105	Drive train: malfunctional; misalignment, unusual noise	Visual and functional
106	Transfer case assembly: malfunction; unusaul noise; leaks, (see 6.3) mounting improper	Visual and functional
107	Compressed air devices: malfunction; unusual noise; leaks	Visual and functional
108	Bilge pumps: malfunction; unusual noise	Functional
109	Cooling system and components: malfunction; unusual noise; leaks; damage; improper clearance	Visual and functional

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

Table III. Classification of Quality Defects

Defects No.	Defects	Method of Inspection
Major	A.Q.L. = 25 Defects/100 Units (cont'd)	
110	Governor linkage and engine governor: malfunction; adjustment	Visual and functional
111	Fuel system components: malfunction; damage; leaks (see 6.3)	Visual and functional
112	Instrumentation switches, warning, indicating and safety devices: malfunction; damage	Visual and functional
113	Electrical system and components: malfunction; damaged light connectors	Visual and functional
114	Service, handbrakes, and locks: malfunction; unusual noise; leaks	Visual and functional
115	Windshield: fogged; cracked; broken; folding interference	Visual and functional
116	Tire: damage; not properly inflated	Visual
117	Lubrication system components: (engine) damage; leaks (see 6.3)	Visual and functional
118	Body, sheet metal, hardware, storage compartments, brackets: improper fits; assembly or installation; defective weldment	Visual
119	Tanks structure and welding defect	Visual
120	Fuel dispensing system: malfunction; leaks (see 6.3); improper assembly	Visual and functional
121	Exhaust system components: damage; leaks	Visual
122	Air cleaner: damage	Visual
123	Seat suspension system components: damage; leaks	Visual and functional
124	Controls: malfunction; clearance improper	Visual and functional

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

Table III. Classification of Quality Defects

Defects No.	Defects	Method of Inspection
Major	A.Q.L. = 25 Defects/100 Units (cont'd)	
125	Winch and wire rope: malfunction; improper size; damage	Visual and functional
126	Walkway coating: application; improper coverage; location	Visual
127	Miscellaneous items or accessories; malfunction; damage	Visual and functional
Minor	A.Q.L. = 150 Defects/100 Units	
201	Controls: malfunction, adjustment, damage	Visual and functional
202	Coolant: low or improper mixture	Visual and hydrometer
203	Lubricants: levels and types improper (all units)	Visual
204	Wheels and tires: improper size; type and mounting	Visual
205	Pulleys and fan: misalignment; improper mounting or clearance	Visual
206	Bolts: defective, loose	Visual and functional
207	Wiring or tubing: defective; improper assembly or assembly	Visual
208	Brake system components: improper assembly installation, or protection	Visual
209	Body, doors, access ladder, access covers, ventilators, stowage brackets, tie downs, lashing hooks and lifting devices: improper fits; assembly or installation; defective weldments; bows; flooring; seals, hardware	Visual and functional

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:23Z
Check the source to verify that this is the current version before use.

Table III. Classification of Quality Defects

Defects No.	Defects	Methor of Inspection
Minor	A.Q.L. = 25 Defects/100 Units (cont'd)	
210	Fuel dispensing system: improper adjustments	Visual and functional
211	Sheet metal and welding defects	Visual
212	Electrical system components: improper installation or assembly	Visual
213	Paint: application and color improper	Visual
214	Steering system components: improper installation of clearance	Visual and functional
215	Exhaust system components: improper assembly or installation	Visual
216	Cooling system components: improper assembly or installation	Visual
217	Fuel system components: improper assembly or installation	Visual
218	Lube fittings: defective; missing or improperly installed	Visual
219	Insulation and noise suppression: missing; improper application	Visual
220	Mud flaps: improper assembly; damage	Visual
221	Walkway coating: excessive application	Visual
222	Seat suspension system components: improper installation or assembly	Visual
223	Cushions, seats, straps, canvas tops, and curtains: damaged; improper fit and installation	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

Table III. Classification of Quality Defects

Defects No.	Defects	Method of Inspection
Minor	A.Q.L. = 25 Defects/100 Units (cont'd)	
224	Miscellaneous items or accessories: missing; proper assembly or installation	Visual and functional
225	Decals, marking, data and instruction plates: incomplete data; missing, improper location or size; vehicle registration and identification marking	Visual
226	Protective coatings: application; missing or properly processed	Visual
227	Lubrication improper	Visual

4.3.1.3 Unclassified defects. All defects having no bearing on function, safety, interchangeability or life, but are considered departures from good workmanship will be noted in writing. Workmanship deficiencies falling within this category and recurring in 5 consecutive lots or ten lots or more within a thirty-day period will be added to the minor classification with no increase to A.Q.L.'s.

4.3.1.4 Recurring major deficiencies. A major deficiency (see 6.8) is recurring when the same defect occurs more than once in the same sample or when the defect occurs in two successive samples. A major defect may be considered recurring when the historical inspection records ("P" chart or approved equivalent) reflect such a condition. Recurring major deficiencies shall be cause for the entire lot or lots to be inspected for the recurring deficiencies. The deficiencies shall be corrected when found.

4.4 Classification of tests.

(a) Acceptance tests, 4.4.1 through 4.4.1.20.
(b) Control tests, 4.4.2 through 4.4.2.9.
(c) Comparison tests, 4.4.3 through 4.4.3.12.

4.4.1 Acceptance test. Each complete vehicle shall be subjected to a 15 mile road test by the contractor over hard-surfaced roads with a grade not greater than ± 1/2 percent. Vehicles (less payload) shall comply to the performance requirements specified herein. Each vehicle shall be operated at speeds specified in 3.5.2 and operated in a reverse gear for a minimum of 100 feet 3 times and shall meet all tests specified at place of manufacture for acceptance test (see table IV).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

4.4.1.1 Test failure. If a vehicle fails to pass any acceptance test specified herein, the Government inspector shall stop acceptance of subsequent vehicles until evidence has been provided by the contractor that corrective action has been accomplished.

Table IV. Location of tests

	Requirement	Paragraph	Place of Mfr.	Proving Ground
Preproduction vehicle	3.1	4.2.1	X	
Initial production vehicle	3.2	4.2.2		X
Reliability	3.2.1	4.2.2.1		X
Durability	3.2.2	4.2.2.2		X
Maintainability	3.2.3	4.2.2.3		X
In-process (inspection)		4.2.2.4	X	X
Completed vehicle inspection		4.2.2.5	X	X
Repair of defects		4.2.2.6	X	
Vehicle disposition		4.2.2.7	X	
Initial production test		4.2.2.8		X
Test failure		4.2.2.8.1		X
Examinations		4.3.1	X	
Construction and acceptance tests	3.4	4.4.1	X	X
Power steering unit	3.4.4	4.4.1.2	X	X
Engine & engine air inductive system check	3.4.5 3.4.5.1	4.4.1.3	X	X
Transmission check	3.4.6	4.4.1.4	X	X
Exhaust system check	3.4.7	4.4.1.5	X	X
Fuel tank & engine stop switch	3.4.8 & 3.4.9	4.4.1.6	X	X

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:23Z
Check the source to verify that this is the current version before use.

Table IV. Location of tests (cont'd)

	Requirement	Paragraph	Place of Mfr.	Proving Ground
Fuel, air, hydraulic lines & fittings check	3.4.10	4.4.1.7	X	X
Governor control linkage check	3.4.11	4.4.1.8	X	X
Personnel heater check	3.4.12	4.4.1.9	X	X
Front winch check	3.4.13	4.4.1.10	X	X
Tank shell check	3.4.14	4.4.1.11	X	X
Emergency valve check	3.4.14.1	4.4.1.11.1	X	X
Piping and accessories check	3.4.14.2	4.4.1.11.2	X	X
Fuel dispensing pump filter separator & valving check	3.4.14.3 & 3.4.14.3.1	4.4.1.11.3	X	X
Reels and hose check	3.4.14.4 thru 3.4.14.4.3	4.4.1.11.4	X	X
Equipment operators cabinet check	3.4.14.5	4.4.1.11.5	X	X
Controls, adjustment, & stowage provisions check	3.4.15, 3.4.16 & 3.4.17	4.4.1.12	X	X
Welding checks.	3.4.18	4.4.1.13	X	X
Tire check	3.4.20	4.4.1.14	X	X
Break-in run and condition after run	3.5 thru 3.5.2.1	4.4.1.15	X	X
Brake test	3.6.7.1 & 3.6.7.2	4.4.1.16	X	X
Steering & articulation functional test	3.6.8 & 3.6.9	4.4.1.17	X	X
Leakage during swimming test	3.6.10 & 3.6.10.1	4.4.1.18	X	X

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

Table IV. Location of test (cont'd)

	Requirement	Paragraph	Place of Mfr.	Proving Ground
Voltage alternator, lights accessories test	3.6.11.2 & 3.6.11.2	4.4.1.19	X	X
Servicing & adjustment check	3.9	4.4.1.20	X	X
Control tests		4.4.2	X	X
Seals check	3.4.1 thru 3.4.1.1	4.4.2.2	X	X
Special kits check	3.4.2 thru 3.4.2.6	4.4.2.3	X	X
Tire chains check	3.4.3	4.4.2.4	X	X
Front winch test	3.4.13	4.4.2.5	X	X
Fuel discharge rate test	3.4.14.1 thru 3.4.14.4.2	4.4.2.6	X	X
Saft top check	3.4.21	4.4.2.9	X	X
Safety verification	3.4.22	4.4.2.10	X	X
Vehicle operational test		4.4.2.7	X	X
Grade & slope operation test	3.6.2	4.4.2.7.1	X	X
Engine starting on grades & slopes tests	3.6.3	4.4.2.7.2	X	X
Cooling system test	3.6.4	4.4.2.7.3	X	X
Speed tests	3.6.6 & 3.6.6.1	4.4.2.7.4	X	X
Stopping brake test	3.6.7.1 & 3.6.7.2	4.4.2.7.5	X	X
Holding on grade test	3.6.7.1 & 3.6.7.2	4.4.2.7.6	X	X

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

Table IV. Location of tests (cont'd)

	Requirement	Paragraph	Place of Mfr.	Proving Ground
Steering angle and vehicle articulation test	3.6.8 & 3.6.9	4.4.2.7.7	X	X
Bilge pump test	3.6.10.1	4.4.2.7.8	X	X
Fuel system cleanliness test	3.6.12 thru 3.6.12.2	4.4.2.8	X	X
Comparison test		4.4.3		X
Special kits operational tests	3.4.2 thru 3.4.2.6	4.4.3.2		X
Exhaust concentration check	3.4.7	4.4.3.3		X
Front winch operational test	3.4.13	4.4.3.4		X
Transportability check	3.4.19	4.4.3.5		X
Vehicle performance tests	3.6, 3.6.2 & 3.6.5	4.4.3.6		X
Environmental operating conditions check	3.6.1	4.4.3.7		X
Steering angle and vehicle articulation test	3.6.8 & 3.6.9	4.4.3.8		X
Leakage during swimming tests	3.6.10	4.4.3.9		X
Bilge pump test	3.6.10.1	4.4.3.9.1		X
Fuel delivery system cleanliness test	3.6.12 thru 3.6.12.2	4.4.3.10		X
Radio interference suppression	3.7	4.4.3.11		X
Noise, vibration, level tests	3.10	4.4.3.12		X
Preservation, packaging & vehicle processing	5.1	4.5	X	X

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

4.4.1.2 Power steering unit check. To determine conformance to 3.4.4, the power steering unit shall be checked to assure proper assembly and installation.

4.4.1.3 Engine and air induction system check. To determine conformance to 3.4.5 and 3.4.5.1, the engine and engine air induction system shall be checked for proper assembly and installation.

4.4.1.4 Transmission check. To determine conformance to 3.4.6, the transmission shall be checked for proper assembly and installation.

4.4.1.5 Exhaust system check. To determine conformance to 3.4.7, the exhaust system shall be checked for proper installation and for leakage.

4.4.1.6 Fuel tanks, hydraulic reservoir, and engine stop switch check. To determine conformance to 3.4.8 and 3.4.9, the fuel tanks and hydraulic reservoir shall be checked for cleanliness, leaks, chaffing and vibration. The engine stop switch shall be operated and checked for functional requirements.

4.4.1.7 Fuel air, hydraulic lines, and fitting check. To determine conformance to 3.4.10, the lines and fittings shall be checked for leaks and signs of excessive vibration and chaffing that could lead to vehicle failure.

4.4.1.8 Governor control linkage check. To determine conformance to 3.4.11, the throttle pedal shall be depressed to stop position and the engine governor control linkage checked to assure that full rack position has been reached without binding or damage to linkages or governor.

4.4.1.9 Personnel heater check. To determine conformance to 3.4.12, the personnel heater shall be checked for proper assembly and installation.

4.4.1.10 Front winch and wire rope check. To determine conformance to 3.4.13, the winch and wire rope shall be checked for proper assembly and installation.

4.4.1.11 Tank shell test. To determine conformance to 3.4.14, every complete tank with all appurtenance welded thereon shall be tested by an air or hydrostatic pressure of 3 PSIG. Air pressure, if used, shall be held for a period of 6 minutes during which the entire surface of all joints under pressure shall be coated with a solution of soap and water, heavy oil or other material suitable for the purpose, foaming or bubbling of which indicates presence of leaks. Hydrostatic pressure, if used, shall be done by using water or other liquid having a similar viscosity, the temperature of which shall not exceed 100°F during the test and applying pressure as prescribed above, gauged at the top of the tank, at which time all joints under pressure shall be inspected for issuance of liquid indicating leaks. All closures shall be in place while test by either method is made. During these tests, operative relief devices shall be clamped, plugged or otherwise rendered inoperative; such clamps, plugs and similar devices shall be removed immediately after the test is finished.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

Any leakage discovered by either of the above described methods, or by any other method shall be deemed evidence of failure to meet the requirements of this specification. Tanks failing to pass this test shall be suitably repaired and the above described tests continued until no leaks are discovered before any cargo tank is accepted by the Government and put into service. Every cargo tank shall withstand the above tests without undue distortion, evidence of impending failure or failure. Failure to meet this requirement shall be deemed as sufficient cause for rejection under this specification.

4.4.1.11.1 Emergency valve check. To determine conformance to 3.4.14.1, the valve shall be operated and checked to assure valve meets functional requirements.

4.4.1.11.2 Piping and accessories check. To determine conformance to 3.4.14.2, all piping and accessories shall be checked for proper assembly and leaks.

4.4.1.11.3 Fuel dispensing pump, filter separator, and valving check. To determine conformance to 3.4.14.3 and 3.4.14.3.1, pump, separator, and valving shall be checked for proper assembly. A pumping check shall be made except that flow rate measurements are not required.

4.4.1.11.4 Reels and hoses check. To determine conformance to 3.4.14.4 through 3.4.14.4.3, the reels and hoses shall be checked for proper assembly, mounting, grounding, types and sizes.

4.4.1.11.5 Equipment operators cabinet check. To determine conformance to 3.4.14.5, the cabinet shall be checked for proper assembly and sealing to assure the restriction of water entrance into cabinet.

4.4.1.12 Controls and adjustment check. To determine conformance to 3.4.15 and 3.4.16 all controls shall be operated and checked for proper installation and functional requirements.

4.4.1.13 Welding and burning check. To determine conformance to 3.4.18, welding repairs and burning procedures shall be checked for use of proper procedures.

4.4.1.14 Tire check. To determine conformance to 3.4.20, the tires shall be checked for proper type and pressures.

4.4.1.15 Break-in run and condition after break-in run check. To determine conformance to 3.5, 3.5.1, 3.5.2, and 3.5.2.1, each vehicle less payload, shall be operated as specified and observed for performance requirements and condition after the break-in run.

4.4.1.16 Brake test. The vehicle shall be operated, service and parking brakes checked for meeting functional requirements, except for grade requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

4.4.1.17 Steering and articulation functional test. To determine conformance to 3.6.8 and 3.6.9, the vehicle shall be operated and functional tests made.

4.4.1.18 Leakage during swimming test. To determine conformance to 3.6.10 and 3.6.10.1, the vehicle shall be placed or driven into water of sufficient depth to float vehicle, leakage shall not exceed that specified. Bilge pumps shall be functionally tested.

4.4.1.19 Alternator voltage, lights and accessories check. To determine conformance to 3.6.11.1 and 3.6.11.2, with all lights and accessories functioning, the alternator voltage shall be checked at point specified and voltage shall be as specified. The lights and accessories shall be operated and checked for functional requirements.

4.4.1.20 Servicing and adjustment check. To determine conformance to 3.9, the vehicles shall be checked for conformance to requirements specified therein.

4.4.2 Control tests.

4.4.2.1 Frequency. The Government shall select at random for control test one out of each 40 vehicles submitted or a minimum of one per month if production is less than 40 vehicles per month. Each vehicle selected shall be operated, with designated or simulated payload, for a distance of 50 miles by the contractor. These tests shall be performed in the presence of a Government representative. After completion of tests the doors, control devices, and other functional parts shall be tested for functional requirements.

4.4.2.1.1 Failure. If the vehicle selected fails to pass any of the control tests, the Government inspector shall stop acceptance examination and testing on subsequent vehicles until such time as conditions causing the failure have been remedied. Any defect found during or as a result of the test shall be prima facie evidence that vehicles accepted subsequent to the previous acceptable control test vehicle are similarly defective unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such defects on all vehicles shall be corrected by the contractor at no cost to the Government. The contractor shall correct defects on all vehicles represented by the failed control test vehicle. Another vehicle with corrective actions implemented shall be subjected to the control test.

4.4.2.2 Seals check. To determine conformance to 3.4.1.1 after fording or swimming operations, all bearings subject to water contamination shall be checked. Lubricants shall be drained and examined for water contamination and shall not be more than that specified.

4.4.2.3 Special kits check. To determine conformance to 3.4.2 through 3.4.2.6, one of each 10 samples of a specific kit furnished shall be installed in accordance with applicable drawings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

4.4.2.4 Tire chains check. To determine conformance to 3.4.3, the tire chains shall be installed and vehicle driven for not less than 2 miles, and retention of tightness and clearance in wheel well checked.

4.4.2.5 Front winch and wire rope test. To determine conformance to 3.4.13, the winch and rope shall be tested with rated load and observed for meeting requirements.

4.4.2.6 Fuel discharge rate tests. To determine conformance to 3.4.14.1 through 3.4.14.4.2, the engine shall be operated at specified speeds indicated for specific semitrailers and the discharge rates checked. Clean kerosene or clean solvent shall be used for pumping and leakage tests. If a solvent is used, the solvent must not be injurious to the filter/separator, tank and plumbing and not leave a residue that will contaminate the fuel dispensing system.

4.4.2.7 Vehicle operational test. The vehicle loaded with rated payload including all stowage and EL item (see 3.4.17) shall be tested as specified herein. The testing where applicable may be performed during the 50 mile road test. The contractor shall be permitted to unpackage and repackage all stowage and EL items for test. Troop installed and depot installed EL will be furnished by the Government.

4.4.2.7.1 Grade and slope operation test. To determine conformance to 3.6.2, the vehicle shall be operated on 60 percent grades headed up and down, and on 20 percent side slopes with each side of vehicle up slope.

4.4.2.7.2 Engine starting on grades and side slopes test. To determine conformance to 3.6.3, the vehicle shall be placed on specified grades and slopes and engine stopped then restarted to assure engine meets starting requirements. The fuel and lubrication systems shall function in all positions without failure.

4.4.2.7.3 Cooling system test. To determine conformance to 3.6.4, during the road test the cooling system shall be checked to assure the system meets cooling requirements. The maximum temperatures of the transmission oil, and engine coolant shall be recorded. Engine oil pressure shall also be recorded.

4.4.2.7.4 Speed test. To determine conformance to 3.6.6 and 3.6.6.1, the vehicle loaded with rated payload shall meet the speed requirements.

4.4.2.7.5 Braking test. To determine conformance to 3.6.7.1 and 3.6.7.2, during road test the service and parking brakes shall be checked for functional operation. Stopping distance shall be not more than that specified.

4.4.2.7.6 Holding on grades test. To determine conformance to 3.6.7.1 and 3.6.7.2, the vehicle loaded with rated payload shall be operated on 60 and 40 percent grade as specified, the vehicle shall be stopped and the transmission put in neutral, service and parking brakes shall hold the vehicle on the specified grades.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

4.4.2.7.7 Steering angles and vehicle articulation test. To determine conformance with 3.6.8, the vehicle shall be operated over terrain permitting articulation (roll) between the front and rear units. The vehicle shall be turned to check conformance with the requirement as specified to the right and the left to assure that there is no binding or interference between the front and trailing units under all conditions of steer and roll. To determine conformance with 3.6.9, stops shall be placed to check the roll limitation as specified.

4.4.2.7.8 Bilge pump operational test. To determine conformance to 3.6.10.1, bilge pumps shall be operated during swimming test to determine performance and rate of water removal.

4.4.2.8 Fuel system cleanliness test. To determine conformance to 3.6.12 through 3.6.12.2 with fuel being pumped through the filter/separator system, the fuel quality at any nozzle shall meet requirements specified.

4.4.2.9 Soft top check. To determine conformance to 3.4.21, the top shall be checked for completeness of assembly.

4.4.2.10 Safety verification. To determine conformance with 3.4.22, all exposed parts which are energized electrically shall be located, insulated, fully enclosed or guarded to prevent hazards to operating personnel and equipment functioning. All moving parts, mechanical power transmission devices, pneumatic components and hydraulic components which are of such a nature or so located as to be a hazard to operating or maintenance personnel shall be enclosed or guarded. Protective devices shall not impair vehicle operation functions. Equipment design which creates hazards shall result in a hazard classification as defined in MIL-STD-882. Those classified as catastrophic or critical will be considered as failures.

4.4.3 Comparison test. The Government may select vehicles anytime during a production contract period and subject vehicles to all tests listed on Table IV and Table V to reveal defects of a workmanship or materials nature that may reduce the effective operation of these items in the field and compare existing quality with previous standards. These tests shall be conducted at Government laboratories or proving grounds designated by the contracting officer. Comparison test vehicles shall be combat loaded with all OEM stored in their applicable spaces. Vehicles selected shall not include any vehicles previously tested for conformance to 4.4.2 (Control tests).

4.4.3.1 Test failure. Failure of the test vehicle to meet minimum performance requirements, failure to comply with the requirements of a workmanship or materials nature and failure to manufacture the vehicle in accordance with the technical data revealed during or as a result of this test, shall be cause for rejection of the vehicle. Further, the Government may refuse to continue acceptance of production vehicles until evidence has been provided by the contractor that corrective action has been taken to eliminate the defects. Any

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:23Z
Check the source to verify that this is the currentversion before use.

defects found during or as a result of this test shall be prima facie evidence that all vehicles already accepted prior to completion of tests, are similarly defective unless evidence satisfactory to the Government is furnished by the contractor that they are not similarly deficient. Such defects on all vehicles shall be corrected by the contractor at no cost to the Government.

Table V. Comparison test (3,6000 miles)

Course	Mileage & speeds	Vehicle payload (see 3.6)
Paved surface (20%)	720 miles at speeds varying up to maximum	8 ton
Secondary roads (36%)	1300 miles at speeds applicable to terrain	8 ton
Hilly cross-country (20%)	720 miles at speeds applicable to terrain	8 ton
Level cross-country (20%)	720 miles at speeds applicable to terrain	8 ton
Belgian block (4%)	140 miles	8 ton
Swimming	1 hour	8 ton

NOTE: 10% of the mileage specified for each course shall be performed with towed load (see 3.6.6.1) excluding swimming.

4.4.3.2 Special kits operational tests. To determine conformance to 3.4.2 through 3.4.2.6, when special kits are furnished for testing; they shall be installed on vehicle and tested for meeting performance requirements for the individual kit.

4.4.3.3 Exhaust concentration check. To determine conformance to 3.4.7, the occupied areas of the vehicle shall be checked for concentration of exhaust fumes. Fumes shall not exceed percent specified.

4.4.3.4 Front winch operational test. To determine conformance to 3.4.13, winch shall be operated with rated load and checked for operational requirements and wire rope functional requirements.

4.4.3.5 Transportability check. To determine conformance to 3.4.19, the vehicle shall be checked for transportability requirements

4.4.3.6 Vehicle performance tests. To determine conformance to 3.6, the vehicle with rated payload shall meet all performance requirements and test results recorded.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:23Z
Check the source to verify that this is the currentversion before use.

4.4.3.7 Environmental operating conditions tests. To determine conformance to high and low temperature operation of 3.6.1, the vehicle shall be tested as required to assure that the vehicle meets all environmental requirements.

4.4.3.8 Steering angle and vehicle articulation test. To determine conformance with 3.6.8, the vehicle shall be operated over terrain permitting articulation (roll) between the front and rear units. The vehicle shall be turned to check conformance with the requirements as specified to the right and the left to assure that there is no binding or interference between the front and trailing units under all conditions to steer and roll. To determine conformance with 3.6.9, stops shall be placed to check the roll limitation as specified.

4.4.3.9 Leakage during swimming operational tests. To determine conformance to 3.6.10, the vehicle shall be operated as specified, performance and leakage checked.

4.4.3.9.1 Bilge pump test. To determine conformance to 3.6.10.1, bilge pumps shall be operated during swimming test to checked performance and rate of water removal.

4.4.3.10 Fuel system cleanliness test. To determine conformance to 3.6.12 through 3.6.12.2 with fuel being pumped through the filter/separator system, the fuel quality at any nozzle shall meet requirements specified.

4.4.3.11 Radio interference suppression test. To determine conformance to 3.7, the vehicle and components shall be subjected to the tactical vehicle suppression requirements of specification specified.

4.4.3.12 Noise and vibration level tests. To determine conformance to 3.10, the applicable sound levels and vibration level instrumentation shall be used to assure the vehicle meets requirements.

4.5 Preservation, packaging, and vehicle processing. Material and equipment shall be inspected prior to shipment to assure conformance to the applicable requirements specified in 5.1.

5. PREPARATION FOR DELIVERY

5.1 Vehicle processing. Unless otherwise specified, the vehicle and equipment shall be processed for shipment and storage in accordance with MIL-V-62038 to the extent indicated on the applicable vehicle preservation data sheet or as specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. The vehicle covered by this specification will be used as a distribution vehicle in the field during tactical military assigned mission

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

operations, will be under extreme climatic conditions, and Phase II air lift. Vehicle will be used to refuel simultaneously several tactical vehicles or Army aircraft and to transfer liquid fuel to other bulk fuel dispensors or containers.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction vehicle, if required (see 3.1).
(c) Special kits, if required (see 3.4.2).
(d) Front winch, if required (see 3.4.13).
(e) Soft top, if required (see 3.4.21).
(f) Quantity of initial production test vehicles required (see 4.2.2.8).
(g) Level of preservation, packaging and processing if other than that specified (see 5.1).

6.3 The following definitions shall be used for the classification of defects for leaks:

(a) Weep: - Any non-recurring evidence of fluid beyond the seal.
(b) Seep: - Any recurring evidence of fluid beyond the seal that does not result in formation of a droplet.
(c) Droplet: - Any recurring evidence of fluid beyond the seal that results in formation of a non-falling droplet.
(d) Drip: - Any recurring evidence of fluid beyond the seal where droplet forms and fall.

6.4 Special kits (see 6.2).

11614953 - Gun ring mounting kit.
11614988 - Arctic Kit.
11618590 - Fuel dispensing meter kit.
11624590 - Canopy and frame kit.
11624639 - Filter kit - infra-red.
11631365 - Warning light kit.

6.5 Mission statement. One mission of the M559 Truck Tank, Fuel Servicing, 2500 gallons shall consist of 100 miles of operation under the conditions of load, speed, terrain and environment shown below and performance of the special functions described.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:23Z
Check the source to verify that this is the current version before use.

Terrain	Distance	Operating schedule Range MPH
Hard surface	20 miles	0 - 30
Secondary	40 miles	0 - 30
Level cross-country	16 miles	0 - 20
Level cross-country MUD*	4 miles	0 - 10
Hilly cross-country *	20 miles	0 - 15

Swimming	*	Traverse Water Barrier 5 Minute Operation
60% Grades:	1/10 of 1% of Hilly cross-country operation	W/O towed load
20% Side slopes:	1/10 of 1% of Hilly cross-country operation	W/O towed load

Climatic Extremes

Temperature:	-25° F to 125° F ambient
Altitude:	500' to 5,000'

TRACTOR FUNCTIONS:

Idle:	30 minutes
Starts:	One cold, one hot at two minutes, one hot at 15 minutes after shutdown (cold means coolant & oil at ambient, hot at maximum obtained).
*4-Wheel Drive:	100% Mud & Swim, cross-country as required, at other operation is in 2-wheel drive.
Reverse:	10 operations at 130 feet each.

Front winch (once/10 missions) one overload attempt (110% of rated capacity) followed by recovery of a load at 90% of capacity at 100 feet distance.

FUEL SERVICE FUNCTION:

Operation:	50% of time with a payload of 16,000 pounds, 25% of time with 1600 pounds, and 25% of time at no load.
Pumping:	Selfload and dispense 16,000 pounds of bulk fuel. Filtered fuel through 50 GPM nozzles, one load/unload cycle per mission.

6.6 TYPICAL VEHICLE LIFE OPERATIONAL PROFILE

MODEL	RENEWAL LIFE MILES - HOURS	AVG M.P.H	TOTAL MISSIONS	MISSION MILES - HOURS
M520	25,000 - 1250	20.0	250	100 - 5
M559	18,000 - 1250	14.4	182	100 - 6.9
M553	12,500 - 1250	10.0	250	50 - 5
M877	25,000 - 1250	20.0	250	100 - 5

In actual troop use, this will vary due to environmental, terrain, and troop mission differences.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

6.7 Process average. Sampling may be initiated if the process average value of the first twenty vehicles inspected is less than the A.Q.L. specified in the classification of defects for major and minor defects. Process average shall be computed as follows:

$$\text{Process average} = \frac{\text{Number of defects}}{\text{Number of vehicle inspected}} \times 100$$

If the computed process average exceeds the specified A.Q.L., 100 percent inspection shall be performed and continued until such time that the process average for twenty consecutive vehicles is less than the specified A.Q.L.

6.8 Definitions of recurring major and minor defects. The Government Inspector will verify that a thorough inspection of each vehicle is performed by the contractor not only for the listed characteristics, but also for any other departures from good workmanship. The Government Inspector will assure that all deficiencies encountered furing the inspection are enumerated on the Deficiency Sheet for the vehicle. The defects noted on Deficiency Sheet shall contain sufficient description to enable the Government Inspector and the contractor's representative to classify the deficiency in accordance with the classification of defects of the vehicle specifications and definitions contained in MIL-STD-105. Corrective action shall be twken for recurring deficiencies.

Custodian:
Army - AT

Preparing Activity:
Army - AT

Project No. 2320-A955

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-T-62130B(AT)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:23Z
Check the source to verify that this is the current version before use.